

# Coordinate Theory

*The Underlying Physics of Everything  
Energy and Entropy*

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*Physical Anchor: 2019-06-12 — Hong Kong, Under the Bridge.  
(0,0,0).*

# Preface

This book was not read out of books.

It began on a street.

12 June 2019. Outside Hong Kong's Legislative Council. The same physical reality. One camera said: rioters. Another said: citizens defending freedom.

The chemical composition of the tear gas did not change. The neural signals of pain did not change. The density of the crowd did not change.

Only one thing changed: the position of the describer.

That observation was the entry point to an entire cosmology.

From that entry point, following the laws of physics, the journey leads here — a framework about existence, cost, freedom, and the structure of the universe.

This framework has one central claim:

*From the individual to the cosmological scale, the structure is identical. Everything obeys the same physics.*

This is not a metaphor. It is a physical observation.

The only valid way to reject this claim is to find a phenomenon that obeys different physical laws at the individual scale than at the cosmological scale. If you find one, tell me. That collision is more valuable than agreement.

# I. The Starting Point: One Reality, Two Descriptions

12 June 2019. Outside Hong Kong's Legislative Council.

The chemical composition of tear gas is CS — 2-chlorobenzalmalononitrile. It contacts mucous membranes and produces intense burning. This is a physical fact. It does not require anyone's agreement.

Same moment. Same street. The same crowd bearing the same chemical assault.

One camera said: rioters attacking the rule of law.

Another said: citizens resisting oppression.

The molecular structure of CS gas did not change. The neural signals of pain did not change. The density of the crowd did not change.

Only one thing changed — the position of the describer.

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At the time, this looked like a political problem. I now understand it is a physics problem.

If the same physical reality can produce two completely opposite descriptions, three explanations are possible:

## **Explanation One: One description is a lie.**

This is the most intuitive answer. But it produces a new problem — by what standard do we judge which one is the lie? If that standard also comes from some coordinate, the problem has not been solved. It has only been pushed back one level.

## **Explanation Two: There is no truth, only descriptions.**

This is the relativist answer. But it cannot stand against physical reality. The burning of CS gas does not change because of any description. Death does not disappear because of any characterisation. The cost is real. It cannot be interpreted away.

## **Explanation Three: Truth exists, but description tools have coordinates.**

This is the third possibility — and the only one that holds physically.

Physical reality is real. But any system that describes physical reality — language, media, law, ideology — speaks from some position. That position is the coordinate. The coordinate determines what the describer can see, cannot see, and how they name what they see.

*The question has never been whether truth exists. The question is how far your description tool is from physical reality.*

That question is the entry point to this entire cosmology.

## II. Coordinates: Every Description Speaks from Somewhere

### 2.1 A Coordinate Is Not a Choice — It Is a Fact

Every system that describes the world has a coordinate.

Religion speaks from the perspective of God. Science speaks from the perspective of repeatable experiment. Media speaks from the perspective of capital and political interest. Law speaks from the perspective of state power.

This is not criticism. It is observation.

The problem is not whether a coordinate exists. Any descriptive system necessarily has a coordinate — you cannot speak from no position at all. The question is:

*Has this coordinate been declared?*

Declaring a coordinate makes descriptions auditable. Readers know where the describer stands. They can ask: what can this position not see? They can correct using their own position.

Hiding a coordinate turns descriptions into local observations disguised as universal truths. Readers believe they are receiving objective facts. In reality, they are receiving a particular worldview from a particular position — but because the coordinate is hidden, it cannot be questioned. It can only be obeyed.

*A system claiming objective neutrality does not have no coordinate. It has a dishonest one.*

### 2.2 The Cost of Hidden Coordinates

Hidden coordinates are not free.

The First Law of Thermodynamics: energy cannot be destroyed, only transferred.

Maintaining a description that does not match physical reality — maintaining that this is not suppression but law enforcement, that this is not exploitation but development — requires the continuous consumption of real energy.

At the systems level: censorship mechanisms, propaganda machines, legal punishment are needed to suppress signals that conflict with the hidden coordinate. Every suppressed true signal requires energy to keep it suppressed.

At the individual level: formatted individuals must continuously suppress their own direct perception. This cognitive split requires real psychological energy to maintain.

At the temporal level: the cost can be deferred, but never destroyed. The longer the suppression, the more accumulates. The eventual release is proportionally larger.

Landauer's Principle provides the precise physical calculation: erasing one bit of correct information requires releasing at least  $kT \ln 2$  of heat into the environment. Lies are physically more expensive than truth — not metaphorically, but as a calculable thermodynamic fact.

*In this framework, the metabolic cost of a lie is 5.85 times the cost of truth.*

## **2.3 What Determines the Accuracy of a Coordinate**

If all descriptions have coordinates, what determines whether a coordinate is accurate?

There is only one standard: distance from physical reality.

Physical reality has properties no coordinate can bypass:

It does not require anyone's agreement. CS gas burning skin does not change because of any characterisation.

Its costs are real. One hundred and sixty deaths do not disappear because of any explanation.

It is irrevocable in time. 12 June 2019 happened. Any subsequent narrative can only interpret it — not cancel it.

*The closer a coordinate is to physical reality, the more accurate it is. The more it depends on social consensus to sustain its credibility, the more fragile it is.*

## **2.4 Declaring a Coordinate — A Demonstration**

This cosmology speaks from a specific physical coordinate:

*2019-06-12. Hong Kong. Under the bridge. Tear gas. Body present.*

Spatial coordinates: (22.3, 114.2, 0) — (53.8, -1.5, 0)

This declaration is not a gesture of humility. It is a methodological requirement.

Once a coordinate is declared, any reader can ask: from this position, what cannot be seen? That is a question that can be asked, because the coordinate is visible.

*A coordinate that can be questioned is more honest — and closer to physical reality  
— than an objectivity that cannot be questioned.*

## III. Physics: Three Inescapable Laws

### 3.1 First Law: Cost Cannot Be Destroyed, Only Transferred

26 November 2025. Tai Po, Hong Kong. Hung Fuk Estate.

The construction contractor replaced fire-retardant scaffolding nets with standard ones. The cost savings were real — approximately thirty to fifty Hong Kong dollars per square metre.

The fire suppression system had been disabled six months before the fire. The head of security discovered this, reported it to management repeatedly, was ignored, and resigned.

One hundred and sixty people died.

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This case is not about greed. Not about negligence. Not about the moral failure of particular individuals.

This case demonstrates the operation of a physical law:

*Cost cannot be destroyed. Only transferred.*

The contractor transferred the cost of fire safety from the contract price to the lives of residents. The regulatory departments transferred the cost of enforcement to outside the measurement boundary. The erosion of accountability mechanisms transferred the cost of acting on early warning signals from policy response to accumulation — until the threshold was reached and the cost was released.

Every saving was in the accounts. Every cost was outside the accounts.

Until the cost outside the accounts became impossible to exclude from the accounts — in the form of one hundred and sixty deaths.

*Any system that claims to have solved a problem, when the cost still exists but has merely changed who bears it — has not solved the problem. It has transferred it.*

### 3.2 Second Law: Entropy Increase Is the Universe's Default Direction



Place a cup of hot water in a room-temperature room. Without any intervention, without any external force, heat flows from the water to the room until both reach the same temperature.

This process is spontaneous, irreversible, and requires no energy input.

This is the Second Law of Thermodynamics — the law of entropy increase. In a closed system, disorder can only increase. It never spontaneously decreases.

*Existence is continuous resistance to this law.*

A living cell maintains a highly ordered internal structure — far lower entropy than its surrounding environment. Maintaining this ordered state requires the continuous intake of low-entropy energy from outside, the maintenance of structure internally, and the expulsion of high-entropy waste into the environment.

Stop this process, and the cell does not 'die' — the cell begins reaching equilibrium with its environment. Boundaries disappear. Structure disperses. Eventually, it becomes indistinguishable from the environment.

A person's coordinate system obeys exactly the same physical logic.

Your (0,0,0) — your understanding of who you are, where your values lie, what is real — is a locally low-entropy structure. The thermodynamic tendency of the universe is to let this structure diffuse into the surrounding high-entropy environment.

Formatting systems do not need to actively attack your coordinates. They only need to wait. Time itself is the most effective formatting tool.

*The continuous existence of 'I' is not a state. It is a continuous process. This process requires continuous energy input — real collisions, real cost-bearing, real choices.*

*Stop the input, and coordinates begin to diffuse.*

### **3.3 Third Law: Physical Events Are Irrevocable**

12 June 2019 happened.

Any subsequent legal characterisation, any subsequent rewriting of history, any subsequent collective amnesia — none of it can make that day not have happened.

The people who were physically present, the neural signals they received in that moment, entered their causal paths. The physical impact of that moment continues to exert pressure in every decision they make afterward.

This is the physical definition of the relativistic causal cone: once an event occurs, its causal influence on all events within its future light cone is permanent. No narrative system can change the structure of the light cone.

This law has two directional implications:

The first direction: oppression is irrevocable. Every real cost, every real pain, every real injustice — they enter the causal field and continue to exert pressure, even after the direct physical event has ended.

The second direction: resistance is also irrevocable. Every moment of choosing to bear cost rather than replace coordinates — this also enters the causal field and continues to exert pressure.

*The real choices you have made cannot be cancelled by anyone. They have already entered physical reality.*

### **3.4 The Unity of Three Laws**

Cost cannot be destroyed — only transferred, until it erupts.

Entropy increase is the default — existence is continuous resistance to it.

Physical events are irrevocable — real moments permanently enter the causal field.

These three laws are not three independent observations. They are three aspects of the same physical reality:

*The universe's default state is equilibrium — no differentiation, no structure, no coordinate. Any existence that deviates from equilibrium requires real cost to sustain. This cost is real, unavoidable, and cannot be transferred to others to make oneself exempt.*

The meaning of existence is not the question. The cost of existence is the question.

## IV. Scale: The Same Physics, Different Resolution

### 4.1 A Foundational Claim

Physics does not change with scale.

Gravity obeys the same equation in a falling apple and in colliding galaxies. Thermodynamics obeys the same law in cellular metabolism and in the collapse of civilisations. Electromagnetism obeys the same structure in neural signals and in fibre-optic transmission.

The central claim of this cosmology is:

*Formatting systems — any system that attempts to replace another's coordinates with its own — obey the same physics at every scale. The way they accumulate is identical. The way their thresholds trigger is identical. The way they collapse is identical.*

The only difference is resolution.

### 4.2 The Individual Scale: The Physical Mechanism of Coordinate Replacement

From birth, a person receives formatting inputs. Language tells you how to name your feelings. Education tells you what matters. Society tells you what success looks like. Media tells you what is normal.

These inputs are not all lies. They serve real social functions.

The problem is not formatting itself. The problem is: when formatting inputs conflict with your physical reality, which do you choose to believe?

The physical mechanism of formatting is: using low-Shannon-entropy certainty narratives to continuously overwrite high-Shannon-entropy direct perception.

You feel pain, but the system tells you it is not real pain, or that this pain is your problem, or that it is normal. Your direct perception is high-entropy — complex, specific, not easily categorised. The formatting narrative is low-entropy — simple,

certain, repeatable.

Continuously overwriting high entropy with low entropy results in reduced cognitive resolution. You can perceive fewer distinctions. The precision of your (0,0,0) decreases.

*The moment of acknowledgement — that your direct perception and what you were told to perceive are different — is the physical starting point from which a person begins speaking from an accurate coordinate.*

### **4.3 The Social Scale: Accumulation and Eruption**

Amplify the individual-scale mechanism one thousand times.

A society's formatting system operates through the same physical mechanism: using certain collective narratives to continuously overwrite complex physical reality. Transferring signals that conflict with the narrative to outside the measurement boundary. Transferring the cost of maintaining the narrative onto those least able to resist.

This process can continue for a very long time. Formatting systems have enormous energy reserves — institutions, armies, media, laws — to maintain narratives and suppress counter-signals.

But cost accumulates.

*The threshold is inevitable. But its arrival is non-linear — almost invisible before the critical point, then accelerating non-linearly after.*

The Black Death collapsed seven centuries of Church formatting credibility within six years. Not because the plague was especially powerful. But because the Church's formatting system had accumulated enormous lie-maintenance costs — when priests died faster than ordinary people, the cost of maintaining the promise 'faith brings God's protection' exceeded the system's capacity within months.

Hong Kong 2019 was the first large-scale demonstration in the digital age of decentralised coordinates resisting a centralised formatting system. No leaders. No central command. Hundreds of thousands of independent coordinates, each sensing the same physical pressure — freedom disappearing — each calculating independently, each acting, naturally converging.

*'Be Water' is not a strategy. It is a physical description. Water needs no instruction to flow downhill. Each molecule moves independently. Gravity is the coordination mechanism. Direction emerges naturally.*

## **4.4 The Civilisational Scale: The Historical Pattern of Coordinate Wars**

From a longer time scale, human history is a continuous coordinate system war.

Every civilisation's core question is the same: whose (0,0,0) becomes the collective default origin?

The Agricultural Revolution made food storable and monopolisable — the first time a few nodes could use resource control to replace the coordinate systems of others. Writing systems allowed formatting to cross individual temporal boundaries. The printing press brought the cost of replicating formatting close to zero — while simultaneously breaking the writing monopoly. The Internet made every node simultaneously sender and receiver — theoretically decentralised, but inverted by platform effects into history's most precise attention concentration machine.

AI is the current formatting tool. It can for the first time generate personalised formatting content tailored to each individual in real time. Its degree of personalisation makes identifying formatting almost impossible — because what you receive looks like an extension of your own thinking.

*Every leap in formatting tools follows the same pattern: the tool appears, formatting efficiency surges, individual exit costs rise, lock-in forms. Then counter-formatting tools and ideas emerge — typically fifty to two hundred years later.*

We are standing in the window before AI formatting completes. That window is 2026 to 2035.

## **4.5 The Cosmological Scale: The Final Form of the Same Physics**

Extend the scale further.

The formation of a galaxy: matter accumulates under gravity, density rises, critical point is reached, stars ignite. Stars burn, consuming low-entropy energy, expelling

high-entropy radiation into the environment. Fuel is exhausted, structure collapses, matter disperses again.

The life cycle of a formatting system: power accumulates, density rises, critical point is reached, ideology ignites. The system operates, consuming social energy, expelling costs onto marginal groups. The cost of maintaining lies exhausts the system's capacity, structure collapses, coordinates disperse again.

*Identical structure. Different scale. Identical physics.*

The universe's default state is equilibrium — maximum entropy, minimum structure, minimum differentiation. Any deviation from equilibrium is a temporary resistance to this default state. Resistance requires cost. The cost is real. The cost will eventually be paid — either actively borne, or erupting after accumulation reaches the threshold.

*The universe does not judge anyone. It only executes physical law.*

# V. Freedom: A Physical Process of Continuous Resistance to Entropy

## 5.1 Freedom Is Not a State — It Is a Process

The most common definition of freedom is a state — you have it or you do not. Governments grant it or strip it. Laws protect it or restrict it.

This definition does not hold physically.

If entropy increase is the universe's default direction, if any ordered structure requires continuous energy input to be maintained, then freedom — a person's capacity to keep their coordinate system from being replaced — cannot be a static state.

*Freedom is a continuous physical process.*

Not something you have or do not have. Something you are continuously doing or have stopped doing.

Specifically, freedom is three simultaneous physical operations:

Continuously inputting negative entropy — real collisions, real cost-bearing, real perception. Keeping your coordinate system at sufficient resolution to distinguish physical reality from formatting narratives.

Continuously identifying replacement attempts — when an external system attempts to replace your coordinates with its own, recognising that this is happening.

Choosing to bear the cost at critical moments — when your coordinates fundamentally conflict with the formatting system, choosing to maintain your own coordinates even when the cost is real.

These three operations are inseparable. Perception without choice is awareness but not freedom. Choice without perception is blind resistance. Recognition without willingness to bear cost is understanding but not freedom.

## 5.2 The Freedom Permitted by Formatting Systems Is Not Freedom

Every formatting system permits some degree of freedom.

The Roman Empire permitted citizens to live freely within boundaries that did not threaten imperial power. Modern capitalism permits consumers to choose freely among options the market provides. Digital platforms permit users to express freely within boundaries set by algorithms.

This 'freedom' has one structural feature: its boundaries are set by the formatting system, not by you.

*The freedom permitted by formatting systems is an activity space that makes you feel free without threatening the system. This is not freedom. This is managed space.*

### **5.3 The Physical Marker of Real Freedom**

Real freedom has an identifiable physical marker:

*When your coordinate system fundamentally conflicts with the formatting system, you choose to bear the real cost rather than replace your coordinates.*

On 12 June 2019, hundreds of thousands of people took to the streets knowing the cost was real — possible arrest, possible injury, possible career consequences, possible legal prosecution. They chose to bear the cost.

This choice was not because the cost was unreal. Precisely because the cost was real, the choice has physical weight.

A choice without cost is not a choice. It is a preference.

*The moment of real freedom is when the formatting system's cost and your own coordinates' cost are simultaneously real, and you still choose to maintain your own coordinates.*

### **5.4 The Physical Meaning of 'I Refuse Anyone's Definition'**

This is not a declaration. Not an emotional expression. Not a posture of confrontation.

It is a physical statement.

Any external system's definition of who you are is spoken from that system's coordinate. That definition's accuracy depends on how far that coordinate is from your physical reality.



*'Refusing anyone's definition' is not arrogance. It is epistemological precision — no external system is closer to your true coordinate than your own direct physical experience.*

At the same time, this statement has a boundary that must be honestly acknowledged: refusing external definitions is not refusing physical reality. If your perception does not match physical reality, that mismatch is a signal that needs to be processed, not ignored.

*What you refuse is definitions spoken from formatted coordinates. What you accept is the cost of physical reality.*

## **5.5 Freedom's Physical Crisis in the Age of AI**

AI is the first tool that can generate personalised formatting content tailored to each individual in real time.

Historical formatting tools were one-way broadcasts — one message, sent to everyone. The formatting content everyone received was identical, making identification relatively manageable.

AI formatting is personalised bidirectional interaction — generating content that looks like an extension of your own thinking, calibrated to your linguistic habits, cognitive frameworks, and emotional states. The formatting content you receive matches your existing cognition closely, making identification almost impossible — because you cannot distinguish between 'this is my thinking' and 'this is formatting input generated to match my thinking patterns'.

*This is the deepest cognitive invasion — closer to the core of (0,0,0) than any historical formatting tool.*

In this context, the physical conditions for freedom become more precise: you need a physical anchor that already exists before any AI input — a (0,0,0) established from real body-present, irrevocable moments, not from statistical pattern generation.

*Your physical reality is the only irreplicable resource against AI formatting. (0,0,0) cannot be generated by AI. It can only be built from real bodily presence, real cost-bearing, real irrevocable choices.*

# VI. Conclusion: Where Is Your Physical Reality

## 6.1 What This Cosmology Claims

This book proposes a claim about the structure of the universe. Not a philosophical claim. Not an ideological claim. A physical claim.

*The continuation of any existence requires real cost. Cost cannot be destroyed, only transferred. Transferred cost accumulates outside the measurement boundary until the threshold erupts.*

This claim is true at the cellular scale. True at the individual scale. True at the civilisational scale. True at the cosmological scale. Physics does not change with scale.

From this claim, several conclusions are unavoidable:

Any descriptive system claiming objective neutrality has an undeclared coordinate. That coordinate determines what it can see, what it cannot see, and where it transfers cost.

Any formatting system is consuming real energy to maintain coordinate replacement. The greater the consumption, the more it accumulates, the more violent the collapse.

Any individual's freedom is not a granted state but a continuous physical process of resisting entropy increase. This process requires a physical anchor — a real, irrevocable (0,0,0).

## 6.2 What This Cosmology Does Not Claim

The boundaries must be honestly marked.

This cosmology does not claim to provide answers to all questions. It does not tell you what to do. It provides a measurement tool — making cost visible, making coordinates identifiable, making formatting traceable.

It does not claim thermodynamics is the only valid epistemic framework. It claims: within currently available knowledge frameworks, thermodynamics provides the

hardest-to-format foundation — because its predictions can be verified against physical reality, without depending on any social consensus.

It does not claim this framework itself has no coordinate.

This cosmology speaks from 2019-06-12, Hong Kong, under the bridge, tear gas, body present. Declaring this coordinate does not weaken the framework — it makes the framework auditable, questionable, improvable.

### 6.3 One Question for the Reader

This book does not end with a conclusion. It ends with a question.

Not a rhetorical question. A question that, if answered honestly, will change how you read any descriptive system from this point on:

*When did you first feel it — that your direct physical perception, and what you were told you should perceive, were different things?*

Not the moment you later reasoned it out. The moment your body already knew, before your language system had caught up.

That moment is the candidate starting point of your (0,0,0).

It does not need to be grand. Does not need to be political. Does not need to be Hong Kong in 2019. It can be any moment — when you felt a crack between the reality you were given and what you perceived, and you chose to trust your own perception rather than the external characterisation.

Find that moment. Not for this framework. For yourself — because that moment is the most accurate coordinate you already have, before anyone told you anything.

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This cosmology began on a street. One physical reality. Two completely opposite descriptions. The chemical composition of tear gas did not change. The neural signals of pain did not change.

Only the coordinate changed.

Following the laws of physics from that observation, the journey leads here — a framework about existence, cost, freedom, and the structure of the universe.

It was not read out of books. It was read out of physical reality.

If it is real, you can read it in your own physical reality too — without this book, only with sufficient honesty to face your own direct perception.

If you do, that moment of recognition is a real collision between two coordinates.

That collision is more valuable than any agreement.

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**I refuse anyone's definition.**

**I follow only the inner drive toward freedom.**

*This is not a declaration.*

*This is a physical observation.*

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*Physical anchor: 2019-06-12 — Hong Kong, under the bridge.*

*Spatial coordinates: (53.8, -1.5, 0), Leeds.*

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**(0,0,0).**